2026-10-27

SiteWhy

Blocked or not, and why

SiteWhy tells a data center developer whether a US site is blocked — and proves why, with the verified rule, the math and what would change it. Track: The Agent That Can Explain Why. Solo team SiteWhy.

Data center moratoria and the site-selection bottleneck

321	**Moratoria or restrictions** Sep 2026, Electric Choice
32	**States with limits** Sep 2026, Electric Choice
NY Executive Order 62	**Statewide moratorium** Jul 2026

Table 1. Moratorium counts, Electric Choice, September 2026; New York Executive Order 62, July 2026

Site selectors and developers triage US greenfield sites before spending on legal review and interconnection queues.

Listing tools (Acres.com, Cherre DataCenter Intel, LightBox, Transect, C&W Athena) list rules and scores but do not reason about whether a rule applies to a site.

SiteWhy answers the harder question: which rules apply to this site at this MW, with the quote, the threshold math, and the flips that would change the verdict.

Source: README problem statement; judging evidence impact.md

Verdicts with the rule, the math and the flip



Figure 1. SiteWhy brand visual: site, grid and rule layers over the US | Source: branding/sitewhy-team.png

- **Verdict:** go, conditional or no-go for a US address plus IT load, with the verified rule text, the applicability math, and what would change the answer.
- **Track:** The Agent That Can Explain Why: every verdict traces to a quote plus arithmetic, never to a completion alone.
- **Scope:** federal Title V, a 1,291-entry moratorium dataset, Texas and ERCOT state and grid rules from curated YAML, county sources (draft ordinance, press release, news article) with verbatim-verified quotes.
- A screening tool, not legal advice.

Source: README; samples/*.json

Agent workflow as a loop, not a chain

1

Plan

the planner drafts targeted county investigation tasks on Nemotron Ultra (planning) · Input: site, MW, jurisdiction match, prior critic objections · Output: per-rule questions for the county investigator; federal, state and grid rules come from curated YAML evaluated in code

2

Investigate

the county investigator reads primary sources · Lightning extraction only, Markovian chunk chain; quotes verbatim · Output: facts with verbatim quotes and source locators

3

Verify

deterministic verbatim quote matching in code, retry at most twice · Unverifiable claims become needs_confirmation, never fatal verdicts · Output: verified, not_found or unsupported per claim

4

Decide

Python applicability decides yes or no; the adjudicator drafts the verdict · The adjudicator synthesizes on Nemotron Ultra · Output: draft verdict with the why-chain and the flips table

5

Critic and trace

red-team objections (critique) loop back to the planner for round 2 · New investigator and verifier passes target the objections; every message is a typed event in the World Model log · Unresolved objections become needs_confirmation; UI lanes show the model and receipt per request

- Real loop in Washington County MD: the critic rejected the draft six-month expiry, round 2 re-investigated, and the expiry ships as needs_confirmation.

Source: md-washington-county-unincorporated.json (events, catches, needs_confirmation[0]); app/orchestrator.py

Technical stack

Layer	Choice
App	FastAPI and uvicorn; SAMPLE_MODE runs the real agents on cached model responses, sources and geocodes, no keys, Zetaris or Nightshift needed
Data	Zetaris SITEWHY tables; DuckDB serves the same SQL when unreachable
AI provider	Meterless Nightshift for every model call, with a per-request receipt
Models	Nemotron Lightning 30B for county extraction; Nemotron Ultra 550B for planning, synthesis and critique
Engines	World Model provenance log; Markovian county reasoning loop

Table 2. Stack as shipped, October 2026

- One run at a time; a run past its deadline returns a partial needs_confirmation report instead of hanging.
- No local GPU needed; Python 3.12 with uv.
- Programmatic entry: python run.py --input input_examples/<site>.json --sample, with per-event traces in logs/*.jsonl.
- Run with Docker: docker build -t sitewhy . && docker run -p 8000:8000 sitewhy.

Source: README; app/llm.py STEP_MODELS; app/data.py; run.py

Zetaris and Meterless integration

Runtime tables	USL artifact rows	USL and quality queries
moratoria plus state_legislation	1,291	3 plus 1 PASS
queried by the agents at run time	usl_site_screen from 4 seed tables, scripted	12 exploratory plus 3 USL queries; null_join_keys 0 violations

Table 3. app/screen.py, app/orchestrator.py; zetaris/usl.sql and zetaris/evidence/, committed f0f2b9b

- Zetaris is the data layer: the agents query SITEWHY.moratoria and SITEWHY.state_legislation at run time; DuckDB serves the same SQL when Zetaris is unreachable.
- The USL is a separate scripted artifact, materialized because plain views do not persist on this instance — evidence the same joins hold, not yet on the agent path.
- Meterless Nightshift is the primary AI provider; model and compute receipt per request appear in the UI agent lanes.
- World Model shared provenance log with a per-run JSONL audit; Markovian chunked county loop, conformance-tested, with 0 tokens saved on current single-chunk sources.
- Cursor: Dockerfile written in a Cursor Agent CLI session (30d5b90); rest built with Claude Code.

Source: app/screen.py; app/orchestrator.py; zetaris/evidence/registration.json, run.json, dq_rule.json

Deterministic checks between model calls

Rule-listing tools	SiteWhy
List rules and scores for an area	Models propose candidate rules; Python decides applies yes or no
Leave the applicability question to the reader	Each claim carries the verbatim quote, source URL and as-of date
Keep no record of which model said what	Computed flips show what would change the verdict; the critic re-investigates

- Models propose, deterministic code disposes, so every verdict traces to a quote plus arithmetic.

Source: README; judging evidence why.md and tech.md

Evaluation and honest limits

Ground truth sites	Seeded critic faults	Recorded samples	Test suite
3 / 3	3 / 3	5	171 passed
Washington MD, Abilene TX, Tucson AZ	fatal-pima, title-v, washington-exemption	4 at 300 MW plus Abilene at 50 MW	pytest, 27 Oct 2026

Table 4. data/eval/results.json; samples/*.json; pytest run 2026-10-27

- Markovian saved 0 tokens on current sources; the engine and its accounting are implemented and tested.
- Seed rules are curated with runtime-verified quotes; autonomous rule discovery is roadmap.
- Coverage is modelled for Texas and ERCOT plus two counties; other states get a federal-only report with an explicit scope note.
- Sample mode replays recorded model answers while the agent logic runs for real.

Source: data/eval/results.json; README limitations; app/llm.py cache replay

A no-go with its verified rule

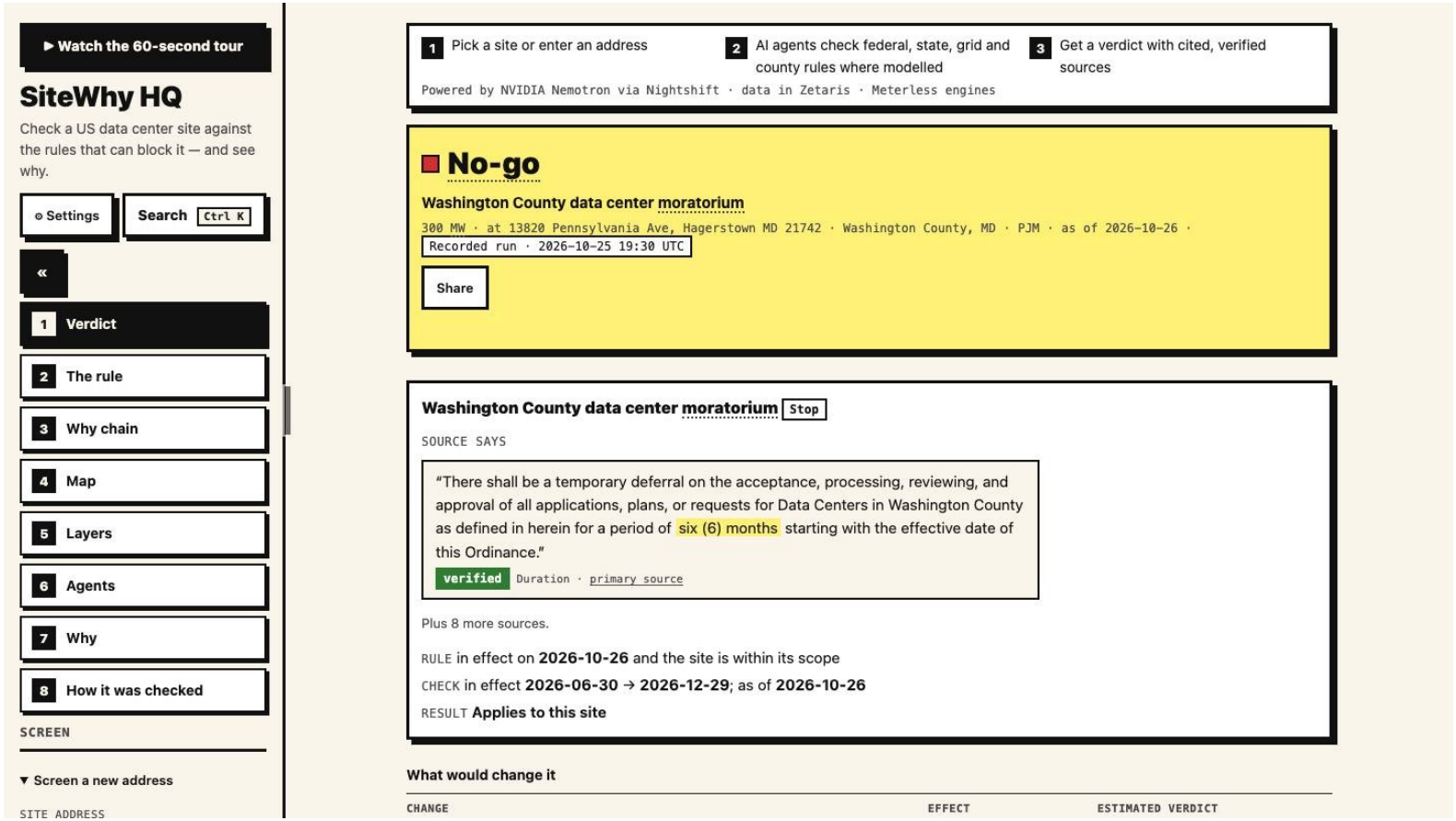
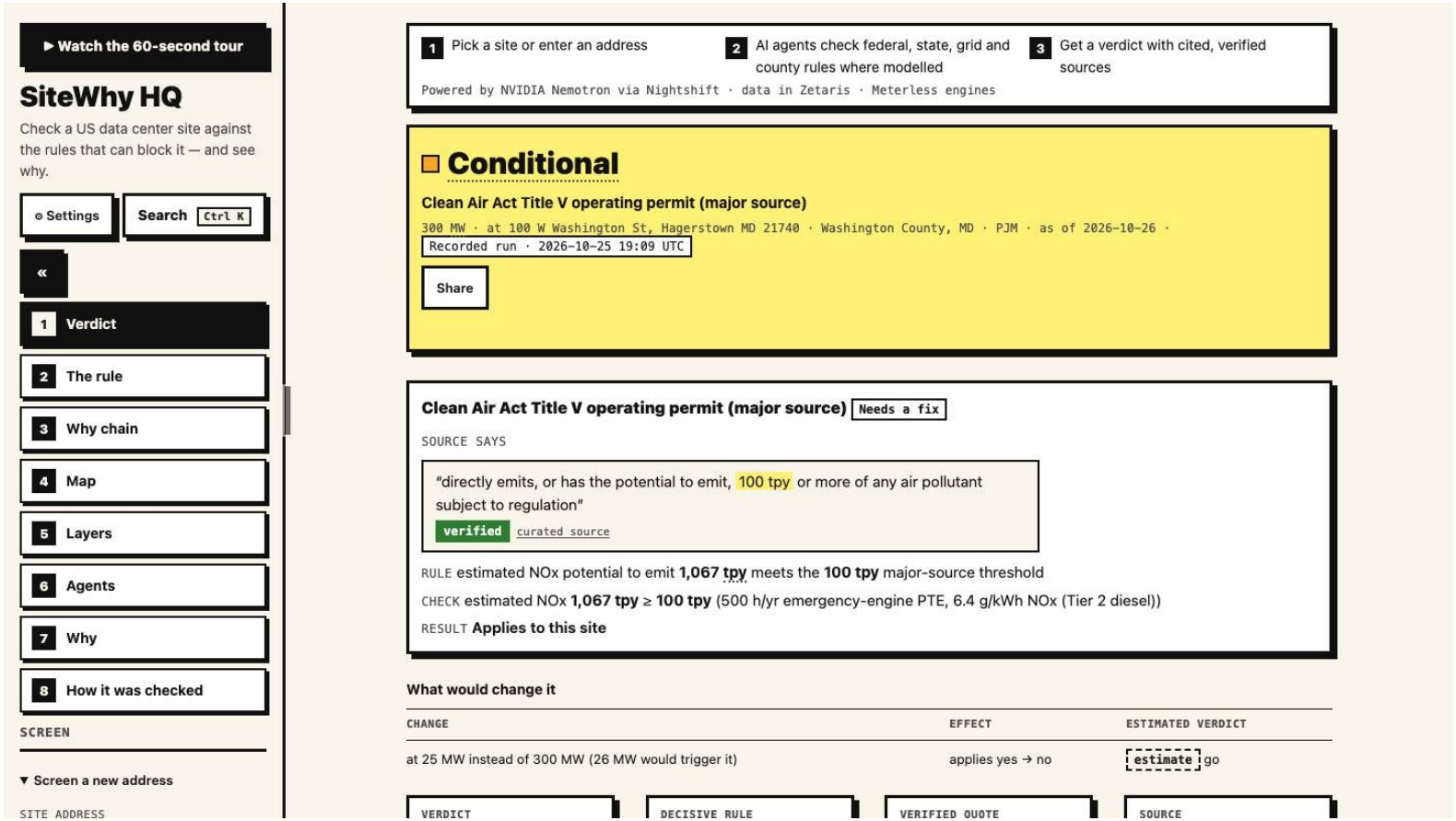


Figure 2. Recorded no-go report: county moratorium with verified quote and applicability math | Source: app screenshot, 27 Oct 2026, 1440×810

- **Finding:** Washington County MD unincorporated at 300 MW is no-go under county moratorium ORD-2026-12.
- **Evidence:** verbatim quote with a verified badge, duration and primary source, then the in-effect check and applies-to-site result.
- **Badge:** showcase reports carry a Recorded run mark with the UTC date; live runs carry none.

Source: samples/md-washington-county-unincorporated.json

Incorporated city versus unincorporated county

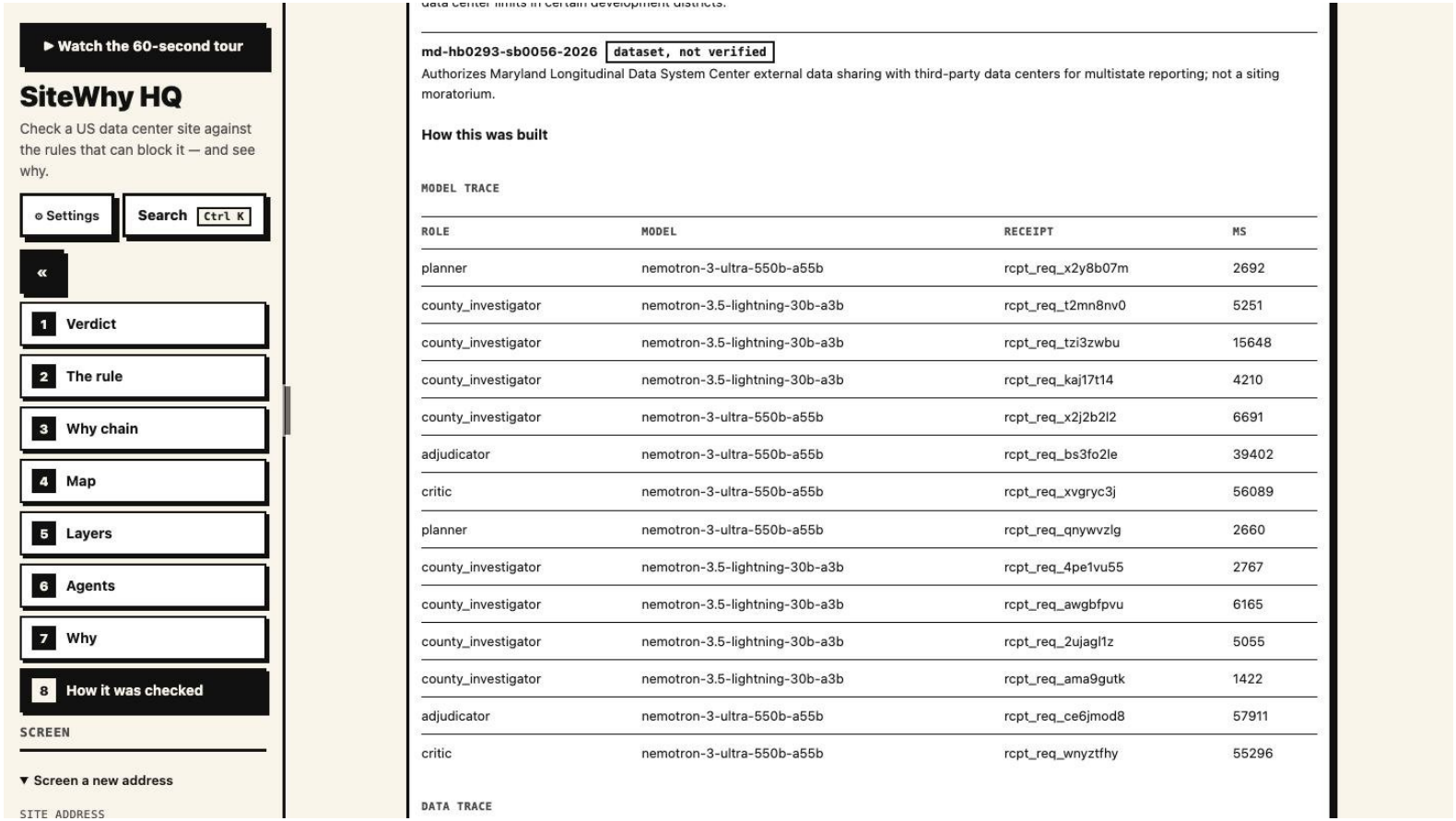


- **Flip:** the Hagerstown city site is conditional because the county ordinance does not cover the city.
- **Math:** the flips table shows that at 25 MW instead of 300 MW the Title V finding changes yes to no, with estimate badges on computed lines.
- The scope flip is the headline the demo leads with.

Figure 3. Recorded conditional report: the same county gives a different answer inside city limits | Source: app screenshot, 27 Oct 2026, 1440×810

Source: samples/md-hagerstown-city.json

Every model call on the record



ROLE	MODEL	RECEIPT	MS
planner	nemotron-3-ultra-550b-a55b	rcpt_req_x2y8b07m	2692
county_investigator	nemotron-3.5-lightning-30b-a3b	rcpt_req_t2mn8nv0	5251
county_investigator	nemotron-3.5-lightning-30b-a3b	rcpt_req_tzi3zwbu	15648
county_investigator	nemotron-3.5-lightning-30b-a3b	rcpt_req_kaj17t14	4210
county_investigator	nemotron-3-ultra-550b-a55b	rcpt_req_x2j2b2l2	6691
adjudicator	nemotron-3-ultra-550b-a55b	rcpt_req_bs3fo2le	39402
critic	nemotron-3-ultra-550b-a55b	rcpt_req_xvgryc3j	56089
planner	nemotron-3-ultra-550b-a55b	rcpt_req_qnywvzlg	2660
county_investigator	nemotron-3.5-lightning-30b-a3b	rcpt_req_4pe1vu55	2767
county_investigator	nemotron-3.5-lightning-30b-a3b	rcpt_req_awgbfpvu	6165
county_investigator	nemotron-3.5-lightning-30b-a3b	rcpt_req_2ujagl1z	5055
county_investigator	nemotron-3.5-lightning-30b-a3b	rcpt_req_ama9gutk	1422
adjudicator	nemotron-3-ultra-550b-a55b	rcpt_req_ce6jmod8	57911
critic	nemotron-3-ultra-550b-a55b	rcpt_req_wnyztfhy	55296

Figure 4. Model trace: role, Nemotron model, compute receipt and latency per call | Source: app screenshot, 27 Oct 2026, 1440×810

- **Trace:** each row names the agent role, the Nemotron model, the compute receipt and the latency.
- **Roles:** Lightning handles county extraction; code verifies quotes verbatim; Ultra handles planning, synthesis and critique.
- 30 traced calls across 5 samples: 18 Ultra and 12 Lightning.

Source: samples/*.json model_trace; README evaluation

Recorded samples and verdicts

Sample	Input	Verdict
md-washington-county-unincorporated	13820 Pennsylvania Ave, Hagerstown MD 21742, 300 MW	no-go
md-hagerstown-city	100 W Washington St, Hagerstown MD 21740, 300 MW	conditional
tx-abilene-ercot	555 Walnut St, Abilene TX 79601, 300 MW	conditional
tx-abilene-ercot-50mw	Same Abilene address, 50 MW	conditional
az-tucson-pima	255 W Alameda St, Tucson AZ 85701, 300 MW	conditional

Table 5. samples/*.json, recorded 25 Oct 2026

- Four sites at 300 MW plus a 50 MW contrast at the same address; verdicts match the data/eval ground truth on the 3 checked sites.
- Abilene 300 vs 50 MW: Title V yes both (1,067 to 185 tpy NOx); SB 6 yes to no; ERCOT audit yes to no; verdict conditional both.
- Each sample carries model_trace with the model per request, plus markovian_runs and events.

Source: samples/*.json (Title V computed.nox_tpy; SB 6 and audit applies fields)

Planned enhancements and current scope

Enhancement	Why it matters
Agent rule discovery	Replace curated seed rules with runtime-verified discovery
More states and ISO rules	Extend beyond Texas and ERCOT plus two counties
Quote-to-value alignment check	Verify the numbers, not just quote presence
Multi-chunk Markovian demo	Show token savings on larger sources
Dockerfile judge path	One-command sample-mode run for judges

Table 6. Roadmap as of 27 Oct 2026; none of this changes the recorded verdicts above

- Current scope stays honest: no accuracy claims beyond the measured 3/3 ground truth and 3/3 seeded faults.
- Pre-window datasets and the Meterless specs are declared as pre-existing third-party material.

Source: plans/m4-submission.md; README limitations

The track and the team

Track: The Agent That Can Explain Why. Solo team SiteWhy. Every verdict traces to a quote plus arithmetic.